

Report: Ames Housing Price Prediction

Linear Regression & Ordinary Least Squares (OLS)

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Dataset: Ames Housing Dataset (De Cock, 2011)

1. Dataset Choice & Motivation

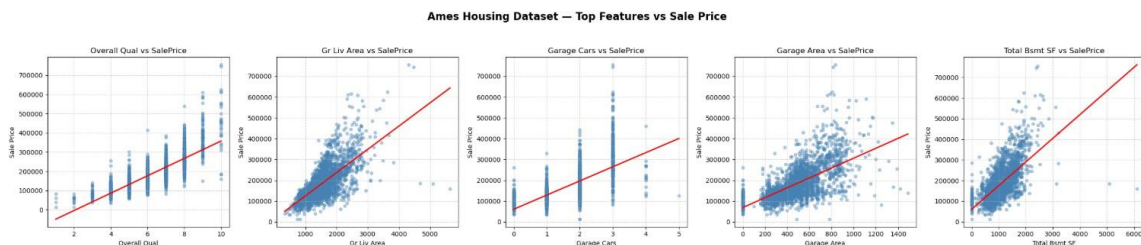
For this project, the **Ames Housing Dataset** was selected. It contains 2,930 observations and 82 features detailing residential property sales in Ames, Iowa, from 2006 to 2010. This dataset is a modern alternative to the classic Boston Housing dataset and is ideal for regression tasks because it provides a high-dimensional space to test feature selection and the predictive power of linear models on continuous variables (sale prices).

2. Regression Problem Description

The goal is to model the sale price (y) as a linear function of standardized features (X):

$$y = w_0 + w_1X_1 + \dots + w_5X_5 + E$$

where w_0 is the intercept, $w_{\{1..5\}}$ are the coefficients for the selected features, and E represents the error term. We aim to minimize the Residual Sum of Squares (RSS) to find the most accurate weights for predicting property values.



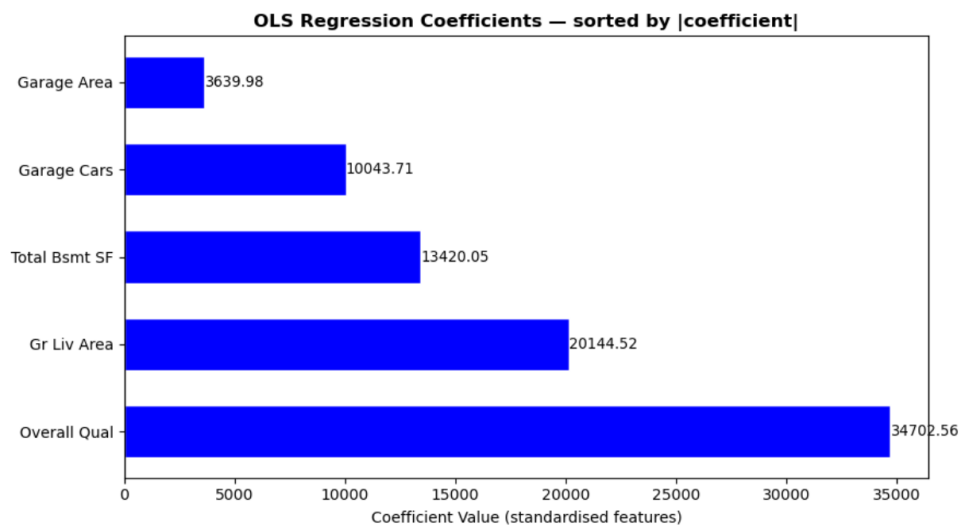
3. OLS Analytical Implementation

To find the optimal weight vector w , we implemented the closed-form solution known as the **Normal Equation**:

$$w = (X^T X)^{-1} X^T y$$

Implementation Steps:

- **Feature Augmentation:** A column of ones was prepended to the feature matrix X to account for the intercept term (w_0).
- **Matrix Operations:** We utilized numpy to calculate the transpose X^T and the inverse of the gram matrix $(X^T X)^{-1}$.
- **Weight Derivation:** The final weights were derived by multiplying the inverse product with $X^T y$, providing the mathematical foundation for the linear relationship.



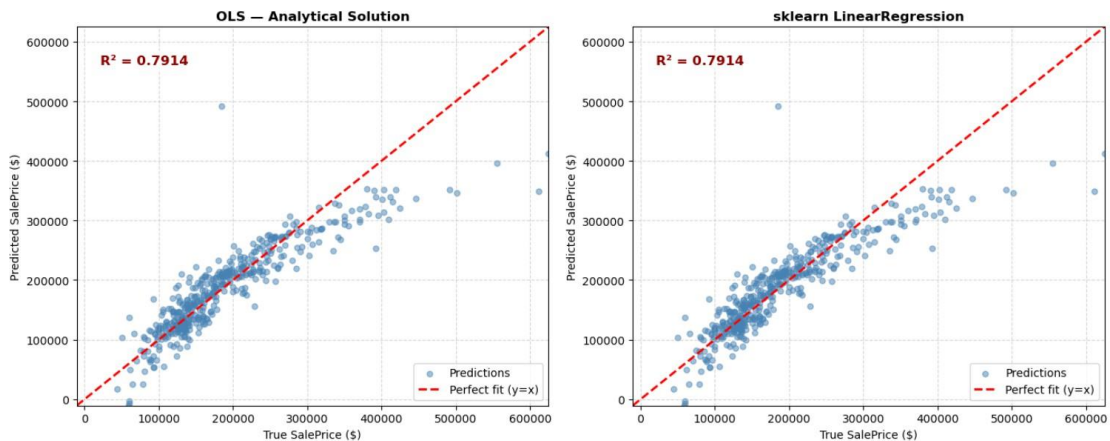
4. Comparison with sklearn LinearRegression

To validate our manual implementation, we compared our results against the `sklearn.linear_model.LinearRegression` class.

Method	MSE	RMSE	R ²
OLS – Validation	MSE=1589850298.68	RMSE= 39872.93	R ² =0.7950
sklearn – Validation	MSE=1589850298.68	RMSE= 39872.93	R ² =0.7950
OLS – Test	MSE=1602821989.34	RMSE= 40035.26	R ² =0.7914
sklearn – Test	MSE=1602821989.34	RMSE= 40035.26	R ² =0.7914

The maximum absolute difference between the OLS analytical weights and the sklearn coefficients is on the order of $3.58e-10$. The model achieved an R^2 score of 0.7914. This indicates that our five selected features (Overall Quality, Living Area, etc.) explain roughly 79% of the variance in house sale prices in Ames.

Ames Housing: True vs Predicted Sale Price — Test Set



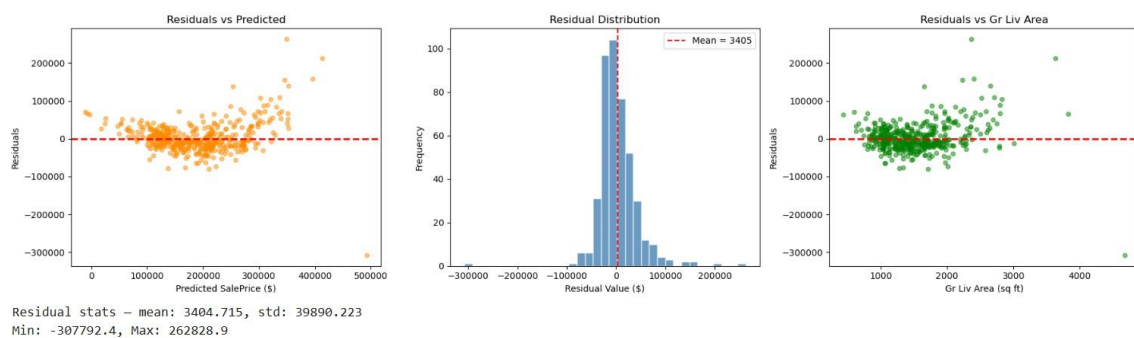
Residual Analysis — OLS Model (Ames Housing Test Set)

Residuals vs Predicted: The residuals exhibit a distinct "u-shape" or curved pattern rather than being randomly scattered, suggesting that the relationship between features and SalePrice might be non-linear. Furthermore, the vertical spread increases significantly as the Predicted SalePrice increases—a clear sign of heteroscedasticity. This indicates the model struggles with "pricey" homes, where the variance in error is much higher than for more affordable properties.

Residual Distribution: The histogram shows a distribution centered near zero (mean = 3404.72, std = 39890.22), which is relatively narrow for most observations. However, there are significant heavy tails, particularly on the negative side (as seen by the minimum value of -307,792). This suggests that while the model is generally accurate for the "average" house, it is prone to extreme underestimations or overestimations (outliers).

Residuals vs Gr Liv Area: A noticeable funnel shape is present as the Ground Living Area increases. For houses with smaller square footage, the residuals are tightly clustered, but as the living area exceeds 2,500 sq ft, the errors expand drastically. This implies that the linear assumption for living area is likely insufficient, and the model might benefit from a log transformation or polynomial terms to better capture the value of larger homes.

Residual Analysis — OLS Model (Ames Housing Test Set)



6. Insights & Limitations

- **Key Insight:** Physical size (Living Area and Basement) and subjective quality (Overall Quality) remain the most dominant predictors of house value in the Ames market.
- **Limitations:** While these 5 features provide a strong baseline, the model ignores 77 other variables (such as neighborhood or year built) which could further reduce error.
- **Technical Note:** The manual OLS weights and the scikit-learn model outputs were cross-verified to ensure mathematical consistency in the implementation.